

Lesson 19 More on Relations

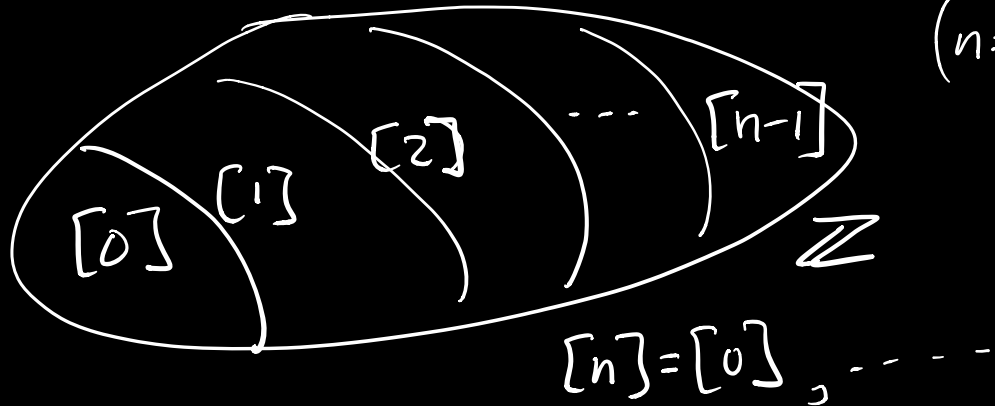
- 1) Continue : equiv rel, congruence mod n
- 2) New type of rel's : partial order

Equiv rel R on a set A : reflexive,
symmetric, transitive

Big thm

An equiv rel on a set A "partitions"
 A into disjoint equivalence classes
(subsets of A) "families"

Congruence mod n , $x \equiv y \pmod{n}$ on $\mathbb{Z}$
 $(n=5)$



n distinct classes:

$$\mathbb{Z}_n = \{ [0], [1], [2], \dots, [n-1] \}$$

$\uparrow$ set

$\mathbb{Z}_n$: new number system w/ only
 n "numbers" : $0, 1, 2, \dots, n-1$

$n=5$: $0, 1, 2, 3, 4$

Modular arithmetic $(+, -, \times, \div)$

$$n=5 : 3+4 = 7 = 2$$

$$7 \equiv 2 \pmod{5}$$

$$\text{Exercise : } [a] + [b] = [a+b]$$

If $a \equiv b \pmod{n}$ and $c \equiv d \pmod{n}$,
 then $a \pm c \equiv b \pm d \pmod{n}$

If $[a] = [b]$ and $[c] = [d]$,

then $[a \pm c] = [b \pm d]$

$$[ac] = [bd]$$

Division is more complicated:
 optional exercise

Def A partial order R on a set A is
 a relation on A such that:

- a) xRx for all $x \in A$ (reflexive)
- b) xRy and $yRz \rightarrow xRz$ ^{$(\forall x, y, z \in A)$} (transitive)
- c) ^(new) if xRy and yRx , then $x=y$ $(\forall x, y \in A)$
 (antisymmetric)

A partial order is called a total order if
 in addition,

- d) ^(new) xRy or yRx for all $x, y \in A$

The most basic example of partial order is " $\leq$ " on $\mathbb{Z}$

(The most basic ex of equiv rel is " $=$ ")

" $\leq$ " is reflexive ($x \leq x$), transitive:

$$x \leq y, y \leq z \rightarrow x \leq y \leq z \rightarrow x \leq z$$

antisymmetric: if $x \overset{(xRy)}{\leq} y$ and $y \overset{(yRx)}{\leq} x$, then

indeed $x = y$

Is it a total order?

Yes, for any integers x, y ,
we have either $x \leq y$ or $y \leq x$.

" $=$ " on $\mathbb{Z}$:

reflexive, transitive

anti-symmetric: if $x \overset{(xRy)}{=} y$ and $y \overset{(yRx)}{=} x$,
then $x = y$ indeed

" $=$ " is a partial order on $\mathbb{Z}$

Is it a total order?

(No): $1, 2 \in \mathbb{Z}$
 $1 \neq 2$ and $2 \neq 1$